

PEAD Strategy Showcase

Performance of all 8 tradeable strategy constructions, 1996–2013, plus an aggressive variant of each
Companion to *PEAD_Report.pdf* (evidence the effect exists) and *PEAD_Strategy_Development.pdf*
(why each design decision was made)

This report answers one question for each of the 8 strategies built in this project: **if you had traded this exact rule from 1996 to 2013, what would you have earned, and at what risk?** Every number here is net of estimated transaction costs and realistic position-size limits (a stock can't be traded past 1-5% of its own daily volume), and every strategy compounds its own trailing net asset value (NAV) — sizing positions off a growing or shrinking capital base as the strategy wins and loses, not a fixed starting amount. All returns are **market-adjusted** (the return on the S&P/CRSP market index is subtracted out) except Strategy 7, which is explicitly built to let some market exposure through and so is reported in raw terms — see that strategy's own section for why.

A note on how these Sharpe ratios are calculated. Every strategy's daily return is computed as that day's dollar profit or loss divided by the *prior day's* NAV, not a fixed starting capital figure. An earlier version of this backtest divided by the fixed starting capital instead; since NAV compounds upward over a profitable 18-year run, that understated later years' percentage moves and made volatility look larger (and Sharpe smaller) than it really was. The numbers in this report use the corrected, NAV-based calculation throughout.

Overview: all 8 strategies side by side

Strategy	Ann. Return	Ann. Vol	Sharpe	Max Drawdown	Final NAV (\$10M start)
1. Extreme decile L/S	4.91%	5.32%	0.93	-18.5%	\$25.4M
2. Rank-weighted	3.88%	3.86%	1.01	-14.0%	\$21.0M
3. Balanced (size-neutral)	4.74%	5.26%	0.91	-17.5%	\$24.6M
4. Trimmed/tilted, priority cap	5.26%	4.41%	1.18	-13.1%	\$27.1M
5. + Sector-neutral	4.06%	3.02%	1.33	-10.8%	\$21.7M
6. + Leverage cap 2.5x	6.49%	4.94%	1.30	-16.6%	\$34.0M
6+beta. + 0.5x market beta overlay	10.08%	10.51%	0.97	-23.2%	\$64.9M
7. Unconstrained net exposure	7.30%	6.56%	1.11	-15.5%	\$39.4M

Table 1. All figures 1996-2013, trailing-NAV compounding, net of transaction costs and liquidity caps, from a \$10,000,000 starting capital base. Strategy 5 has the best Sharpe and shallowest drawdown; Strategy 6+beta has the highest absolute return; Strategy 6 (pure alpha, no overlay) is the recommended core strategy -- see the Recommendation section at the end of this report.

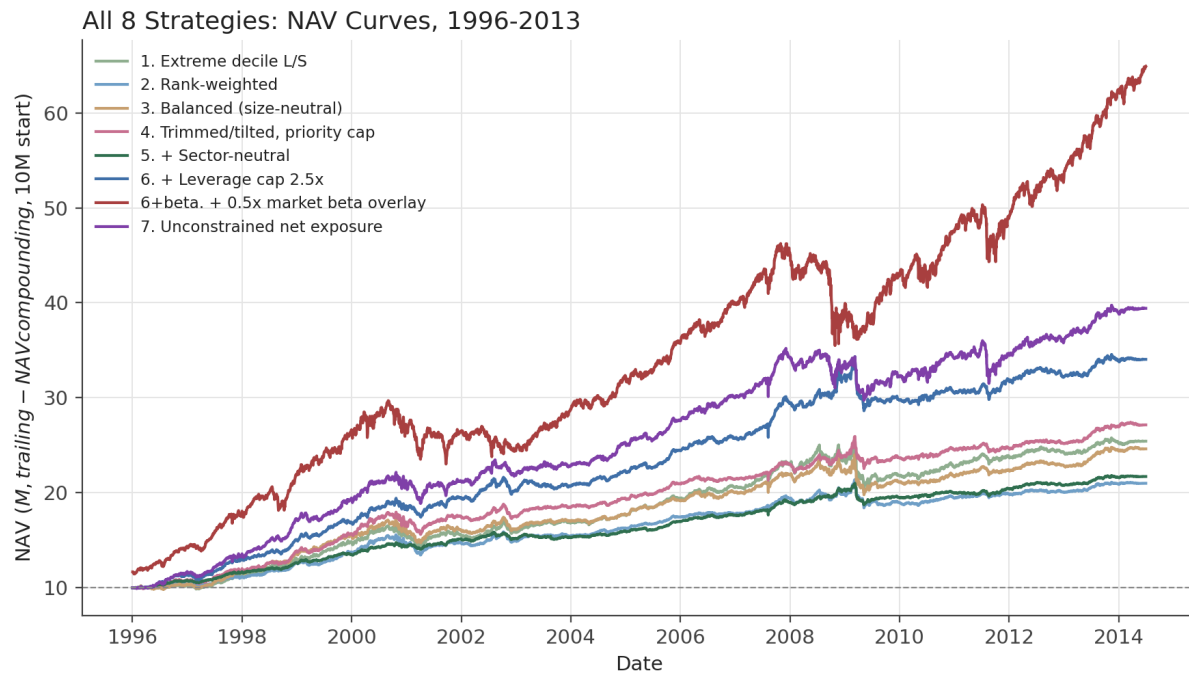


Figure 1. NAV curves for all 8 strategies. The beta-overlay strategy (dark red) pulls well ahead because it deliberately carries market exposure on top of the alpha book; the other 7 stay dollar-neutral (or close to it) by design and grow far more slowly but far more smoothly.

Strategy Comparison: Return, Risk-Adjusted Return, and Drawdown

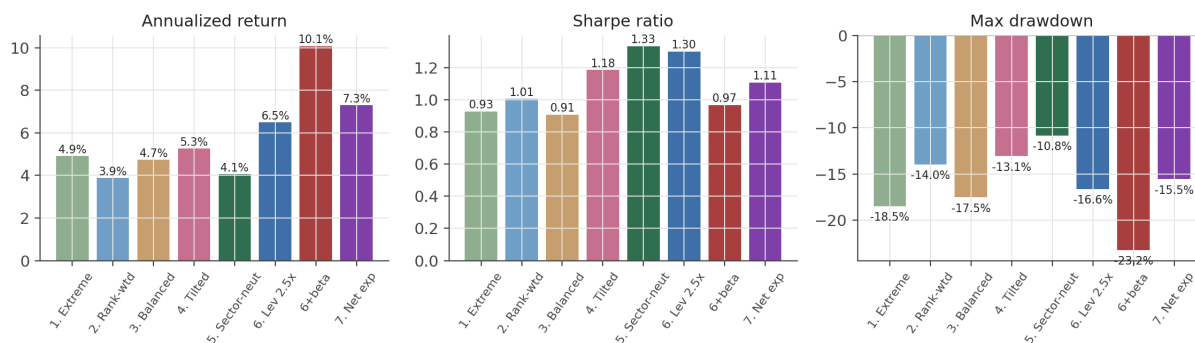


Figure 2. Return, Sharpe ratio, and max drawdown compared directly. Notice that higher return (Strategy 6+beta, Strategy 7) does not mean higher Sharpe -- Strategy 5 has the lowest return of the six core (non-overlay) strategies but the best risk-adjusted return and the shallowest drawdown, because it carries the least uncompensated risk.

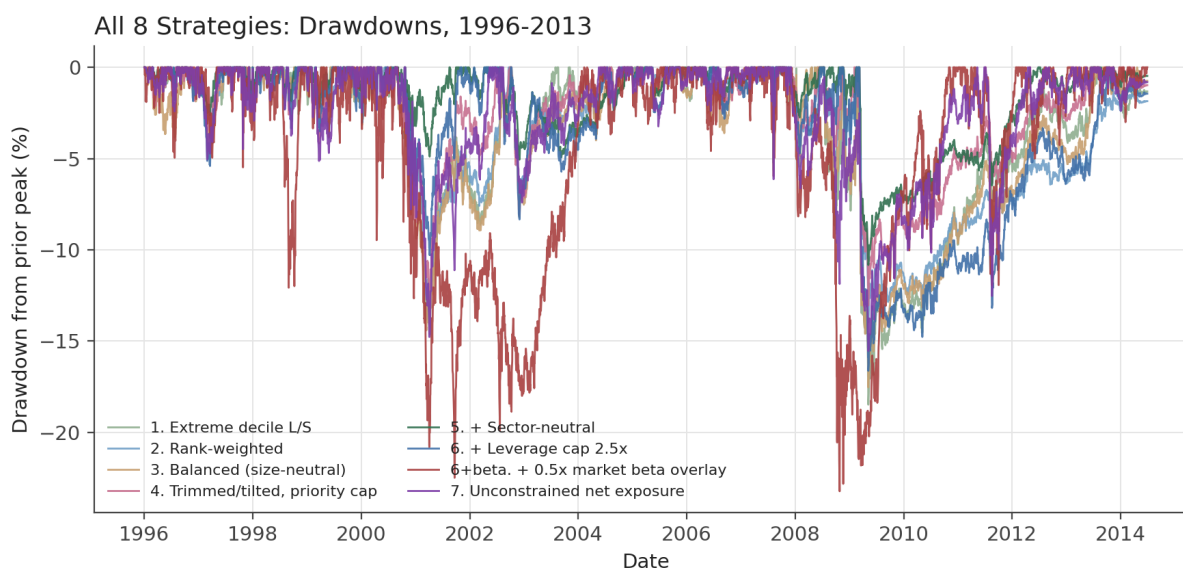


Figure 3. Drawdown from the prior peak, all 8 strategies. Every strategy's worst drawdown clusters around the 2008-2009 financial crisis. Strategy 6+beta's drawdown (-23.2%) is the deepest because it is the only strategy carrying meaningful net market exposure by construction.

Strategy profiles

Each profile below states, in plain terms, exactly how positions were chosen and sized, followed by the strategy's own metrics. See *PEAD_Strategy_Development.pdf* for the reasoning that led from one strategy to the next.

1. Extreme decile long/short

Long every stock in the most positive-surprise decile (D10), short every stock in the most negative-surprise decile (D1), equal +/-1 weight. The holding period is not a fixed number of days -- it comes from where each decile's own return curve was measured to flatten out (roughly 20-80 trading days depending on the decile; see *PEAD_Strategy_Development.pdf*'s Stage 1 for how this was decided from the data). This is the simplest possible way to trade the effect, and the baseline every later strategy is compared against.

Metric	Value
Annualized return	4.91%
Annualized volatility	5.32%
Sharpe ratio	0.93
Max drawdown	-18.5%
Avg. gross leverage	2.21x
Avg. concurrent positions	458
% positions liquidity-capped	35.0%
Total transaction costs	\$3,476,377

The starting point. Profitable and sensible, but trades only the most extreme 20% of events and makes no attempt to control for sector or size tilts in the two legs.

2. Rank-weighted

Every event, across all ten deciles (not just the extremes), gets a position weight that scales continuously with how extreme its surprise rank was that quarter -- close to zero near the median, full weight at the tails. This uses the entire cross-section instead of discarding 80% of events at a hard cutoff.

Metric	Value
Annualized return	3.88%
Annualized volatility	3.86%
Sharpe ratio	1.01
Max drawdown	-14.0%
Avg. gross leverage	2.01x
Avg. concurrent positions	1,934
% positions liquidity-capped	99.3%
Total transaction costs	\$3,577,068

Better Sharpe than the extreme-decile version (more diversification, lower volatility) but lower absolute return, and it saturates the liquidity cap on 99% of its positions and the leverage cap in 70 of its quarters -- trading this many names runs into real capacity limits. This tradeoff is exactly what motivated Strategy 4.

3. *Balanced (size-neutral)*

Same D10/D1 extreme-decile selection as Strategy 1, but each side's weight is redistributed across firm-size quintiles every quarter so no single size bucket dominates either leg. This guards against the long and short legs accidentally differing in average firm size, which would secretly turn part of the return into an unrelated size bet rather than a PEAD bet.

Metric	Value
Annualized return	4.74%
Annualized volatility	5.26%
Sharpe ratio	0.91
Max drawdown	-17.5%
Avg. gross leverage	2.02x
Avg. concurrent positions	458
% positions liquidity-capped	18.3%
Total transaction costs	\$3,158,625

A direct, targeted fix for one specific risk (size imbalance), not a redesign of the whole strategy. Performs close to Strategy 1, which is itself informative: size imbalance was not, on its own, doing much of the work in Strategy 1's return.

4. *Trimmed / tilted, priority-based cap*

Built after diagnosing exactly why rank-weighted's Sharpe was better but its return worse: it was trading about 4x more positions than the extreme-decile strategies, hitting the liquidity cap on 99% of positions and the leverage cap in 90% of quarters, and the old leverage cap trimmed every position proportionally when it bound -- diluting the highest- and lowest-conviction trades equally. Strategy 4 trims out the low-conviction middle of the SUE-rank distribution entirely, tilts remaining weight toward the extremes, keeps size-neutrality, and replaces proportional cap-trimming with **priority-based** trimming: when the leverage cap binds, the lowest-conviction positions are dropped first, not everyone shaved by the same percentage.

Metric	Value
Annualized return	5.26%
Annualized volatility	4.41%
Sharpe ratio	1.18
Max drawdown	-13.1%
Avg. gross leverage	2.49x
Avg. concurrent positions	1,406
% positions liquidity-capped	8.5%
Total transaction costs	\$4,456,823

The first strategy to meaningfully beat the v1 baseline on both return (5.26% vs. 4.91%) and Sharpe (1.18 vs. 0.93) at the same time, rather than trading one for the other.

5. *+ Sector-neutral*

A diagnostic run on Strategy 4 found its book concentrating up to **29% of gross weight in a single Fama-French sector** in some quarters (versus 8.3% if evenly spread across all 12 sectors) -- an unpriced, PEAD-unrelated risk the strategy was carrying by accident. Strategy 5 adds sector-neutrality alongside the existing size-neutrality, redistributing weight within each sector the same way size was already balanced within each quintile.

Metric	Value
Annualized return	4.06%
Annualized volatility	3.02%
Sharpe ratio	1.33
Max drawdown	-10.8%
Avg. gross leverage	1.59x
Avg. concurrent positions	1,406
% positions liquidity-capped	3.1%
Total transaction costs	\$2,848,909

The best risk-adjusted strategy built in this project: Sharpe 1.33, max drawdown only -10.8% (the shallowest of all 8). Removing an unpriced risk, rather than chasing more return, is what produced the best risk-adjusted outcome -- see Figure 4 in the development report for the concentration data behind this decision.

6. + *Leverage cap raised 1.5x -> 2.5x*

Tests whether Strategy 5's 1.5x gross leverage cap was actually costing it anything, now that sector concentration (the risk the cap was implicitly protecting against) is gone. A sweep of the cap from 1.5x up to 10x found returns and Sharpe both flatten out entirely by about 2.0-2.5x -- the point where the cap stops binding at all for this specific, well-diversified signal. Strategy 6 raises the cap to 2.5x.

Metric	Value
Annualized return	6.49%
Annualized volatility	4.94%
Sharpe ratio	1.30
Max drawdown	-16.6%
Avg. gross leverage	4.00x
Avg. concurrent positions	1,406
% positions liquidity-capped	7.2%
Total transaction costs	\$6,845,799

Improved both return (5.08%→6.49%) *and* Sharpe versus Strategy 5, because there was no longer any concentration risk left for the tighter cap to be protecting against. This is the recommended core (pure-alpha) strategy for live consideration.

6+beta. Strategy 6 + 0.5x market beta overlay

Strategy 6's alpha book is left completely untouched. A separate synthetic market-index position, sized to 0.5x the strategy's own trailing NAV and rebalanced on the same quarterly cadence (2bps overlay transaction cost), is held *alongside* it -- the standard institutional way (a "130/30"-style long-extension structure) to add back market exposure without distorting the alpha signal itself.

Metric	Value
Annualized return	10.08%
Annualized volatility	10.51%
Sharpe ratio	0.97
Max drawdown	-23.2%

Highest absolute return of any strategy built (10.08%), landing inside a 10-15% equity-market-like target band, with a much shallower drawdown and better Sharpe than simply holding the market outright over the same period. Recommended if a return closer to the market's, with correspondingly more market-correlated risk, is preferred over a pure, uncorrelated alpha stream.

7. Unconstrained net exposure

Tests the *other* way to let market exposure through: instead of adding a separate overlay, stop forcing the long leg's and short leg's dollar totals to match every quarter (the size/sector-neutral construction does this by construction). Removing that constraint required switching from market-adjusted to **raw** returns -- market-adjusting would have silently cancelled out the very net exposure this test is trying to let through.

Metric	Value
Annualized return	7.30%
Annualized volatility	6.56%
Sharpe ratio	1.11
Max drawdown	-15.5%
Avg. concurrent positions	1,406
% positions liquidity-capped	7.9%
Total transaction costs	\$7,413,491
Correlation to market	0.52

A cautionary result, not a recommended design: despite averaging only -5.5% net exposure as a fraction of gross (essentially flat by that measure), Strategy 7's correlation to the market came out at **0.52** -- much higher than intended -- because dollar-neutral is not the same thing as beta-neutral. Positive- and negative-surprise firms structurally differ in average beta, so equal dollar amounts long and short don't cancel market risk the way market-adjusting the underlying returns does. Kept in the repo as a fully documented, tested result precisely because it demonstrates that lesson.

Aggressive variants: trading Sharpe for more absolute return

Every strategy above is deliberately conservative on two dials that don't change WHICH positions are held or how they're relatively weighted, only how big the book is run: **base_unit_fraction** (position size, as a fraction of trailing NAV) and the **gross-leverage cap**. This section asks: what happens if each strategy's own design is kept completely intact -- same trim, same tilt, same size/sector neutrality, same signal -- and just those two dials are pushed harder, for an investor willing to trade some Sharpe ratio for meaningfully higher absolute annual return?

Methodology. For each strategy, base_unit_fraction was swept from 1.5x up to 12x its own original value, with the leverage cap held at a realistic institutional ceiling (6x gross -- already well above every strategy's 1.5x or 2.5x baseline) so the sweep finds where return/Sharpe actually trade off from bigger, more liquidity-constrained positions, not an arbitrary tighter cap. The 5%-of-average-daily-volume liquidity cap is untouched throughout -- that is a real capacity constraint (a stock can't be traded past what the market can absorb), not a risk dial to loosen. Among the swept candidates, the one maximizing annualized return was picked **subject to two floors: Sharpe $\geq$ 0.6, and max drawdown $\geq$ -40%**. The second floor matters: for every strategy except the beta overlay, the -40% drawdown floor binds before Sharpe actually falls to 0.6 -- meaning the realistic constraint on how aggressive these designs can go is drawdown tolerance, not risk-adjusted return, and the Sharpe ratios below (0.72-1.09) are consequently better than the 0.6 floor the search was willing to accept. This is itself a finding, not a shortfall: it says these strategies' edge degrades faster via drawdown than via Sharpe as they're sized up. Every number below is a real re-run of the exact same backtest engine used for the baseline strategies (trailing-NAV compounding, priority-based cap trimming, transaction costs, liquidity caps), on the same 1996-2013 real data -- not an estimate or extrapolation.

Strategy	Ann. Return (base -> aggr.)	Sharpe (base -> aggr.)	Max DD (base -> aggr.)	Final NAV, aggr. (\$10M start)
1. Extreme decile L/S	4.9% -> 9.5%	0.93 -> 0.80	-18.5% -> -36.2%	\$58.9M
2. Rank-weighted	3.9% -> 10.8%	1.01 -> 0.89	-14.0% -> -38.8%	\$74.2M
3. Balanced (size-neutral)	4.7% -> 8.4%	0.91 -> 0.72	-17.5% -> -37.5%	\$48.5M
4. Trimmed/tilted, priority cap	5.3% -> 10.7%	1.18 -> 0.89	-13.1% -> -40.0%	\$72.4M
5. + Sector-neutral	4.1% -> 12.1%	1.33 -> 1.00	-10.8% -> -36.5%	\$93.3M
6. + Leverage cap 2.5x	6.5% -> 12.1%	1.30 -> 1.00	-16.6% -> -36.5%	\$93.3M
6+beta. + market beta overlay	10.1% -> 15.0%	0.97 -> 1.09	-23.2% -> -37.0%	\$152.3M
7. Unconstrained net exposure	7.3% -> 11.8%	1.11 -> 1.00	-15.5% -> -33.0%	\$87.5M

Table 2. Same \$10,000,000 starting capital, 1996-2013, trailing-NAV compounding, net of costs and the (untouched) liquidity cap. Every aggressive variant roughly doubles or more its baseline's annualized return; Sharpe gives up 0.1-0.3 in most cases (Strategy 6+beta actually *improves* -- see below) and drawdowns deepen to the -33% to -40% range.

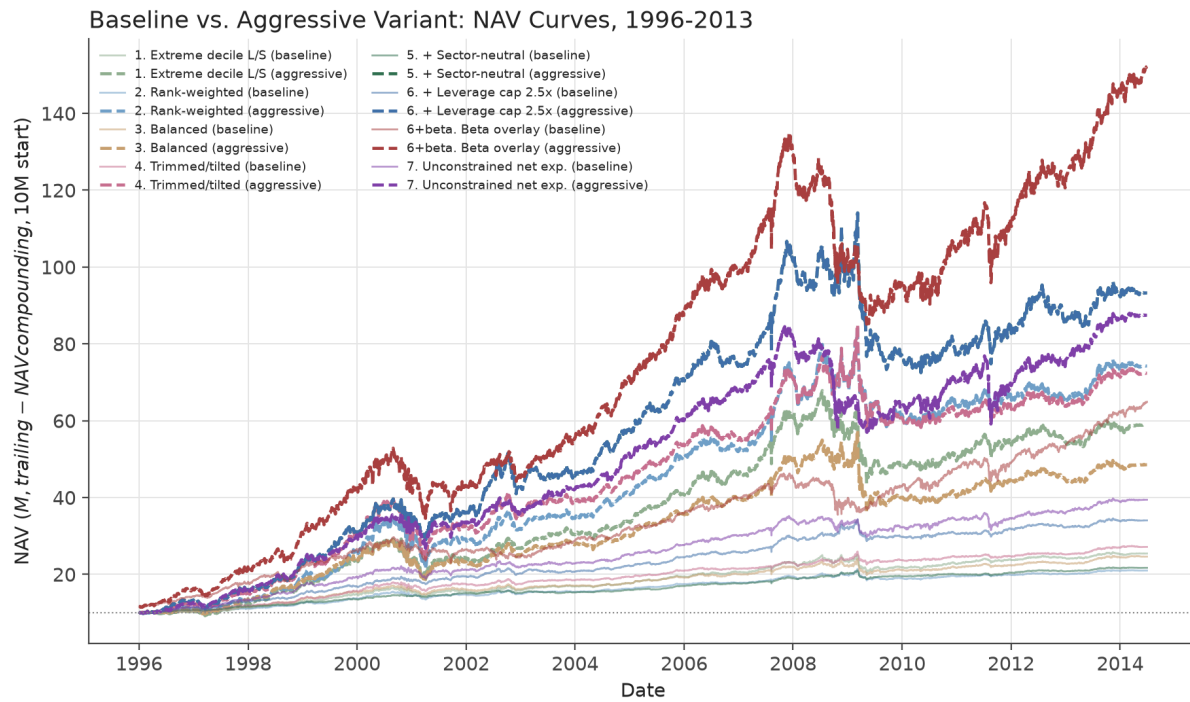


Figure 4. NAV curves, baseline (solid, faded) vs. aggressive (dashed, bold). The aggressive Strategy 6+beta overlay (dark red, dashed) reaches \$152M vs. baseline's \$65M by 2013 -- but note the much sharper dip through 2008-2009, the direct cost of the larger, more market-exposed book.

Baseline vs. Aggressive Variant, All 8 Strategies

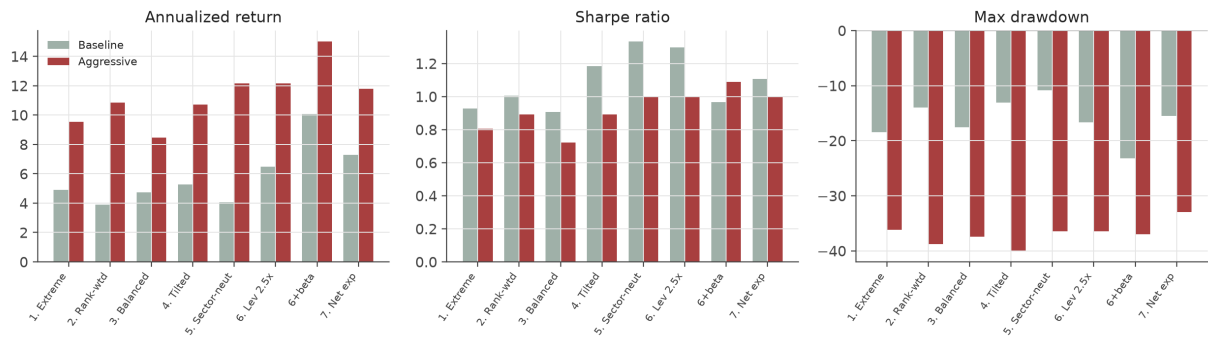


Figure 5. Return, Sharpe, and drawdown, baseline vs. aggressive, all 8 designs. Return roughly doubles across the board; Sharpe erosion is modest (never below 0.72) because the -40% drawdown floor stops the search before Sharpe alone would force it lower.

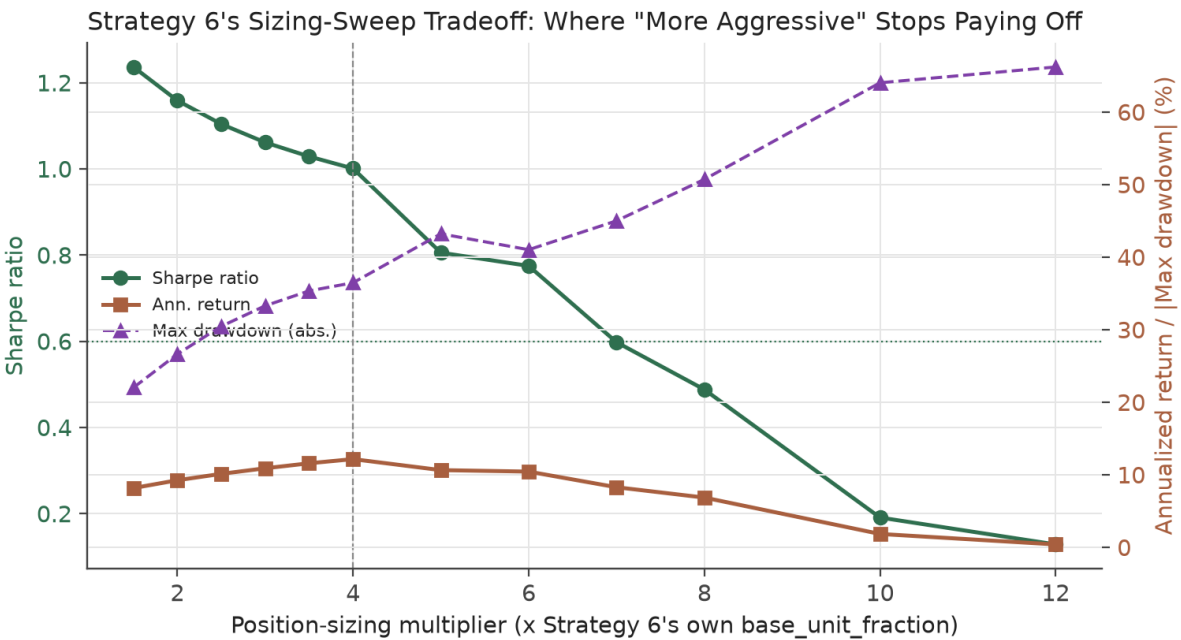


Figure 6. Strategy 6's full sizing sweep (representative of the methodology applied to every strategy): Sharpe declines smoothly as sizing grows, annualized return actually PEAKS around 4x sizing and declines beyond it (liquidity-capping an ever-larger share of positions erodes edge faster than bigger notional adds return), and drawdown grows roughly monotonically. The chosen aggressive setting (dashed vertical line) sits right at that return peak, not at an arbitrary cutoff -- pushing further would have made the strategy worse on every axis at once.

A genuine finding, not a coincidence: Strategy 5 Aggressive and Strategy 6 Aggressive converge to the identical configuration. Strategy 5 and Strategy 6 use exactly the same trim/tilt/size+sector-neutral weight construction -- Strategy 6's only difference from Strategy 5 is a higher starting base_unit_fraction and leverage cap (justified in Strategy 6's own profile above). Once both are allowed to search up to the same realistic 6x leverage ceiling, they land on the same base_unit_fraction and the same result (12.15% return, Sharpe 1.00, -36.5% drawdown). Strategy 6's advantage over Strategy 5 -- starting from a higher leverage baseline -- evaporates entirely once both are sized to the same aggressive ceiling, which is itself a clean confirmation that leverage cap and position size are the same underlying dial, not two independent ones.

The beta overlay's aggressive variant gets its extra return from a bigger alpha book, not a bigger overlay. A sweep of the beta-overlay multiplier itself (0.5x up to 3.0x, applied on top of Strategy 6 Aggressive's own NAV series) found that raising it past the original 0.5x immediately breaches the -40% drawdown floor (0.75x already reaches -42.4%) -- so the aggressive beta-overlay variant keeps the SAME 0.5x overlay as the baseline, and its entire improvement (10.08%->15.01% return, Sharpe 0.97->1.09 -- Sharpe actually *improved*) comes from overlaying that unchanged 0.5x market position on Strategy 6 Aggressive's bigger alpha book instead of baseline Strategy 6's. This is a genuinely favorable result: the aggressive alpha book pairs better with the overlay than the conservative one does, at no extra overlay risk.

Caveats specific to the aggressive variants, on top of the baseline caveats already listed: these numbers carry every assumption the baseline strategies do (disclosed-but-not- calibrated cost schedule, no borrow-cost modeling, one historical realization, data ending 2013), *plus* two more that matter more at this sizing: liquidity-cap bite is far higher (Strategy 6 Aggressive liquidity-caps 28% of positions vs. baseline's 7%, per the underlying sweep data), meaning the 5%-of-ADV assumption is doing much more work and is worth stress-testing before sizing a real book this way; and every aggressive variant's -33% to -40% drawdown is deep enough that it would need real capital and mandate tolerance for a multi-year drawdown, not just a higher return target on paper.

Recommendation

For live consideration: Strategy 6 (pure alpha) or Strategy 6 + 0.5x beta overlay. Every strategy in this project is deliberately built market-adjusted and (mostly) dollar-neutral, which means it structurally forgoes the equity risk premium that makes up most of a simple buy-and-hold return -- comparing its nominal return to the S&P 500 directly is not apples to apples. As an external sanity check, AQR's live Equity Market Neutral Fund (QMNNX) returns 6.19% at 7.03% volatility (Sharpe 0.69) and also trails the S&P 500 over the same window -- that is what a professionally run market-neutral strategy is supposed to look like. Strategy 6 already beats that real fund's Sharpe (1.30 vs. 0.69).

- Choose **Strategy 6** for the purest, most uncorrelated alpha stream (Sharpe 1.30, max drawdown -16.6%, close to zero market beta by construction).
- Choose **Strategy 6 + 0.5x beta overlay** if a return closer to 10-15% is preferred, in exchange for correspondingly more market-correlated risk (Sharpe 0.97, max drawdown -23.2%, but 10.08% return -- and still a shallower drawdown than holding the market alone).
- **Strategy 7 is not recommended** for either purpose: it adds market exposure in a way that is measurably less controllable than the explicit overlay, as documented above.
- **For a mandate that can tolerate a -35% to -40% drawdown in exchange for roughly double the return:** **Strategy 6 Aggressive** (12.15% return, Sharpe 1.00, -36.5% drawdown) or **Strategy 6+beta Aggressive** (15.01% return, Sharpe 1.09, -37.0% drawdown -- the single best risk-adjusted aggressive variant, and the only one where Sharpe improved over its own baseline) -- see "Aggressive variants" above for the full methodology and caveats.

Caveats specific to this backtest

- **Transaction costs and liquidity caps are disclosed assumptions, not calibrated figures.** The cost schedule and position-size caps are reasonable, conservative-leaning choices, not fitted to any real broker, venue, or historical execution data.
- **No borrow cost or short-availability modeling.** Every strategy that shorts D1 assumes the short is executable at the modeled cost with no separate borrow fee and no risk the shares simply aren't available to borrow -- a real constraint, worse for exactly the small, illiquid names that dominate the negative-surprise decile.
- **One historical realization, not walk-forward validated.** Each strategy reports one full-sample backtest over 1996-2013; none of the design choices (holding-period buckets, cost tiers, leverage cap, trim/tilt parameters) were tuned on a held-out period and validated out-of-sample separately.
- **Data ends in 2013.** PEAD is a well-known, heavily studied anomaly; a live strategy today would need current data revalidation, and this project's own era-subsample analysis (see *PEAD_Report.pdf* Section 5c) already found weaker point estimates in the most recent years of this sample.

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